

Replication Guide

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Overview

This document explains how to reproduce the results presented in the report, including both the proposed method and the baseline comparison.

Step 1: Install Dependencies

```
pip install -r requirements.txt
```

Step 2: Verify Files

Ensure the following files exist:

- model/processed_kdd.csv
- model/model_processed_kdd.h5
- fairness_tool.py
- lab4.py
- plot_graph.py

Step 3: Run Proposed Method

```
python fairness_tool.py
```

This generates:

- fairness_results.csv
- fairness_summary.csv
- wilcoxon_results.csv

Step 4: Run Baseline

```
python lab4.py
```

This generates:

- baseline_results.csv
- baseline_summary.csv

Step 5: Generate Graph

```
python plot_graph.py
```

This creates:

```
idi\_vs\_budget.png
```

Expected Results (Proposed Method)

Budget	Mean	Std	Median	Min	Max
100	0.0253	0.0138	0.0200	0.0000	0.0600
300	0.0242	0.0080	0.0233	0.0133	0.0467
500	0.0278	0.0079	0.0260	0.0140	0.0460
1000	0.0262	0.0043	0.0260	0.0180	0.0340

Notes

- Each experiment uses 30 seeds
- Results should be very close to those in the report
- Small variations may occur across systems
- TensorFlow warnings are normal